

INFERENCE-CHANGE ASSURANCE

Bounded inference - change assurance

Determine whether one specific model, provider, or execution-policy change is supported against quality, reliability, latency, and economic requirements declared before the comparison — while preserving the evidence, the uncertainty, and the provenance needed to reconstruct the verdict.

LOCAL-FIRST

NOT A GATEWAY

NOT A GENERIC EVAL PLATFORM

NO PRODUCTION-READINESS CLAIM

NO CUSTOMER-VALIDATION CLAIM

01 · Problem

Migration decisions are commonly made from aggregates over successful requests. That framing erases billable retries and fallbacks, converts missing cost evidence into zero, hides a regressed critical segment inside a favorable mean, and compares fields that share a name without sharing a meaning.

02 · System purpose

InferenceLedger adjudicates one proposed change against predeclared requirements for a defined workload and accounting basis. Routing and execution exist only to collect evidence. Canonical engineering is maintained privately; this public repository is the shareable reference surface.

03 · DECISION PIPELINE · FREEZE → CAPTURE → LEDGER → VERIFY → DECIDE

01 Freeze Workload identity, baseline and candidate, critical segments, required floors, accounting basis. experiment.json	02 Capture Provider attempts, retries and fallbacks, latency, usage, errors, monetary origin. ProviderAttempt	03 Ledger Reconstructable run evidence. Calculated, reported, and UNKNOWN amounts stay distinct. JSONL + SQLite	04 Verify Pairing, coverage, segments, paired statistics, completeness, field comparability. pairing-audit.json	05 Decide Every required check resolved independently; bundle retains inputs and uncertainty. decision.json
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04 · RETAINED EXPERIMENT LEDGER · ZERO APPROVALS RECORDED

EXPERIMENT	OUTCOME	OBSERVED SIGNAL	NOT ESTABLISHED
IFstruct JSON - mode 100/100 pairs · local gemma3:270m	INCONCLUSIVE gate 6 pass · 0 fail · 2 inconclusive	p50 14226 → 9912 ms · structure 0/100 → 0/100 · required cost checks had no variation	Latency improved and approval was still withheld. Both arms failed the structural check, so this is not quality evidence; local cost is not invoice evidence.
Synthetic JSON - mode 32/32 pairs	INCONCLUSIVE relative gain, no approval	exact-field 0/32 → 8/32 · mean successful latency +137.40625 ms	Pre-registration chronology was not independently Git-proven and was not upgraded after execution.
OpenRouter billing 50/50 paid requests · 0 retries	VALIDATED_MISMATCH mismatch retained	4 frozen monetary rules matched exactly · frozen prompt-token rule did not · key delta \$0.00070845	One gateway reconciliation surface. Not direct OpenAI billing-ledger truth, invoice equivalence, or generalized billing correctness.
External pilot · Task 5 preparation only	NOT STARTED no instance frozen	provider / customer calls 0 · authorized paid spend \$0	Customer, commercial, and comparative decision-value validation remain unestablished at this snapshot.

05 · Decision semantics

APPROVE Supported inside the declared evidence boundary.	REJECT A required floor is violated.	ABSTAIN Incomplete, incomparable, or inconclusive.
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The gate is an intersection of required checks, not a weighted score. Committed artifacts keep their original SHIP / NO_GO / REVIEW / INCONCLUSIVE labels; the public set above is a mapping, not a rewrite.

06 · Invariants

- The provider attempt is the economic unit, not the request.
- Unknown cost stays unknown; it is never aggregated as zero.
- Provenance is not comparability.
- A relative gain does not satisfy an absolute floor.
- Completeness of representation is not sufficiency of evidence.
- A frozen decision is not rewritten by later interpretation.

07 · Validation boundary

Mechanisms and committed artifacts are engineering-verified within this snapshot. External, customer, and commercial validation are not established. At the evidence snapshot, Task-5 preparation existed, no concrete pilot instance was frozen, execution had not started, provider and customer calls were 0, and authorized paid spend was \$0. No claim is made that this system outperforms a competent conventional workflow with equivalent evidence access.

08 · Reproduce without a provider

```
$ make check
$ python scripts/audit_decision_completeness.py \
  --decision-json benchmarks/reports/ifstruct-json-mode-20260911/decision.json
```

The audit reports that required evidence surfaces are represented. The same bundle remains INCONCLUSIVE.